Marco **Vetrano**

Data Analyst | PhD in Physics

About me

Data Analyst and **Computational Physicist** with a background in **Machine Learning** and High-Performance Computing (HPC). I have gained experience in managing large amounts of data and modeling complex systems at international research centers. Used to translating technical analyses into clear and accessible insights, I combine scientific rigor and lateral thinking to develop innovative solutions to complex problems.

Area of Specialization

Physics • Machine Learning
Network Analysis • Data Analysis

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Experience

2023–Present

Data Analyst | Quantum Computing Researcher
University of Palermo · Italy

- Development of hybrid algorithms (Quantum-Classical) for the analysis and classification of large astrophysical datasets (PNRR STILES Project).
- Optimization of computational pipelines on HPC infrastructures (Leonardo) and IBM Quantum devices, reducing execution times for complex simulations.
- Stack: Python, HPC.

2025

Data Analyst | Quantum Machine Learning applied to Time Series Predictions
IFISC - University of the Balearic Islands · Spain

- Design and implementation of predictive models (Quantum Reservoir Computing) for the analysis of chaotic time series.
- Application of Reservoir Computing techniques to improve forecast accuracy on nonlinear data.
- Stack: Python, HPC.

Education

2026 **PhD in Physics**
· University of Palermo

2026 - Present **Master in AI Development and Engineering**
ProfessionAI ·

2023 **Master's Degree in Physics**
· University of Palermo

2019 **Bachelor's Degree in Physical Sciences**
· University of Palermo

Languages

Italian	C2	Native speaker
English	C1	● ● ● ● ● ●

Technology Stack

python · SQL · Hugging Face · Pytorch · Langchain · TensorFlow · Git · XGBoost · Docker · Linux · LaTeX · Quarto · AWS · PySpark · FastAPI

Publications

2025 "State estimation with quantum extreme learning machines beyond the scrambling time.", Springer Nature npj Quantum Information 11.1 (2025): 20.

2025 "Exoplanetary atmospheres retrieval via a quantum extreme learning machine.", arXiv preprint arXiv:2509.03617.

Certificates

2026 Udemy certification: Machine Learning and Data Science in Python

2023 Duolingo C1 certification in English

Projects

2025 **QuantumToolBox**
University of Palermo · Italy
A Python package developed mainly using the numpy, scikit-learn, and plotly libraries for implementing quantum dynamics and building classification and time series prediction models. The package is available on GitHub in the QuantumStuff repository

2025-2026 **AstroQELM**
University of Palermo · Italy
A Python package designed for end-to-end analysis of spectral data. The package includes functions for reading and parsing h5 and txt data, preprocessing and compression with scikit-learn, processing and classification with qiskit, numpy, and scikit-learn. The package is available in the AstroQELM repository

2026 **Instagram-Analyzer**
· Italy
Backend app developed with FastAPI in Python for scraping Instagram profiles. The app uses an Instagram username entered by the user, from which it extracts the latest uploaded reel and its associated comments using two Apify actors, and computes BPM and duration. It also includes two Hugging Face AI models for speech-to-text transcription of the video and sentiment analysis of the comments. The app is fully containerized and available on GitHub.

2026 **Olist-App**
· Italy
Project for the end-to-end analysis of the Olist e-commerce dataset, freely downloadable from Kaggle. The package, fully containerized with Docker, contains functions for data download, database construction with SQLAlchemy, querying, aggregation, and data cleaning in PostgreSQL. It also includes three sentiment analysis models of increasing complexity, developed with scikit-learn, PyTorch, and Hugging Face. Currently under development, it will in the future include a graphical interface developed with Streamlit, a customer care agent, and a Tableau interface for visualizing aggregated data with SQL.

Other Experience

2022 and 2024 **Academic Tutor (STEM)**
University of Palermo · Italy

- Teaching support for Physics, Mathematical Analysis, and Geometry courses.
- Personalized mentoring for students with learning disabilities, translating complex mathematical concepts into accessible explanations.

2022 **Data Analyst Intern**
Ri.MED Foundation · Italy
Application of Network Analysis techniques for the study of RNA structures, identifying critical interaction patterns.

2018 **Data Analyst Intern**
IASF - Institute of Space Astrophysics and Cosmic Physics · Italy
Development of scripts in C for the processing and spectral analysis of data from black holes.

Awards

2023 First place at **QuHack4AI: Quantum Hackathon for Industrial Applications**, organized by the University of Bologna and Unipol, with a project on variational algorithms for the study of hydrogen and water molecules

2023 **"Eduardo Gugino" Award** for the master's thesis on "Information Scrambling in Quantum Extreme Learning Machines" for the special March 2023 session.

Talks

2023 "Information Scrambling in Quantum Extreme Learning Machines for State Estimation" at the New Trends in Non-equilibrium Statistical Mechanics 2nd Course

2024 "QELM response to scrambling dynamics and applications to atmospheric retrieval" at the International School on Non-Equilibrium Phenomena - Quantum Reservoir Computing Working Party

2025 "Exoplanetary Retrieval with Quantum Extreme Learning Machines" at the National Congress of the Italian Physical Society

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